Nivedhaa Naresh Kumar

nkmar@bu.edu | LinkedIn | GitHub

EDUCATION

Boston University Expected May 2028

B.S. in Data Science

Relevant Coursework: Multivariable Calculus, Digital Logic & Computer Architecture, Physics: E&M, Intermediate Microeconomics

In Progress: Deep Learning, Data Structures & Algorithms (Rust), Statistics & Linear Algebra, Intro to AI

WORK EXPERIENCE

AI/ML Intern May 2025 – Oct 2025

Emory University School of Medicine – Madabhushi Lab, Atlanta, GA

- Designed and trained CNN and Transformer-based models in PyTorch/TensorFlow, achieving 89% classification accuracy on endometrial ultrasound datasets
- Built end-to-end data pipeline including preprocessing, segmentation, feature engineering, and contrastive learning for model optimization
- Reduced model training time 6x (3+ hrs → 30 min) by optimizing preprocessing and model architecture

Data Engineering Intern May 2025 – Aug 2025

Georgia Tech – Structured Information Neuroengineering Lab, Atlanta, GA

- Developed full-stack React dashboard to collect, structure, and visualize behavioral experiment data
- Designed real-time data ingestion pipeline integrating hardware instrumentation into web-based system
- Enabled cognitive psychologists to conduct studies 4x faster (1 hr+ → 15 min) through streamlined experiment setup

Research Assistant Jan 2025 – Dec 2025

University of Rochester – Photoacoustic & Ultrasonic Research Lab, Rochester, NY

- Queried and cleaned large-scale signal metadata using SQL for downstream statistical analysis
- Co-designed ultrasound probe for real-time fetal brain monitoring to detect birth asphyxia
- Generated visualizations from raw ultrasonic signals to support faster clinical interpretation

PERSONAL PROJECTS

Ultrasound Image Quality Filter Python, Scikit-learn, UMAP, NumPy, Seaborn

- Trained modified ResNet50 (grayscale) classifier with AdamW + cosine LR; extracted penultimate-layer embeddings for analysis
- Applied UMAP and distance-based outlier detection to flag noisy images and visualize dataset structures

Stock Financial Analysis Tool Python, BeautifulSoup, yfinance, Requests

- Engineered automated data pipeline to scrape and structure financial metrics, integrating live market data via yfinance API
- Designed rule-based valuation framework using P/E ratios, revenue growth, and earnings trends

LEADERSHIP & COMMUNITY

Upsilon Pi Epsilon – Boston University Feb 2026 – Present

- Selected for technical honors society competitive rush process and technical interview
- Participated in professional development and networking events within computing community

Chemistry Teaching Assistant Aug 2025 – Dec 2025

University of Rochester

- Led two weekly recitation sections for 30+ students, improving exam scores by 30%
- Graded 120+ exams per cycle and provided detailed performance feedback

TECHNICAL SKILLS

Languages: Python, SQL, Rust, JavaScript

Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, Image Segmentation

Data Analytics & Tools: Pandas, NumPy, Spark, Tableau, Excel, UMAP, Git, Linux, React, HTML/CSS